

Solutions for Problem Set 9

Mathematics for Social Scientists

EXERCISE 1. Solve the following counting problems.

- (i) You have 3 letters addressed to 3 friends, which you deliver one at a time. How many different ways are there of ordering your delivery?
- (ii) The letters are coloured blue, red, and green. If you choose the delivery order at random, what is the probability that the red letter is delivered first?

i. 6

ii. $\frac{1}{3}$

EXERCISE 2. You are dealt 5 cards from a normal deck of 52 cards.

- (i) What is the probability that none of the cards are picture cards?
- (ii) What is the probability that all of the cards are of the same suit?
- (iii) What is the probability that exactly 3 of the cards are kings?

i. $\frac{40 \cdot 39 \cdot 38 \cdot 37 \cdot 36}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}$

ii. $\frac{\binom{13}{5} \binom{4}{1}}{\binom{52}{5}}$

iii. $\frac{\binom{48}{2} \binom{4}{3}}{\binom{52}{5}}$

EXERCISE 3. Every weekday (Mon-Fri) you go to a café for lunch. There are 2 cafés close to the university, and you choose between them randomly each day.

- (i) What is the probability that you don't visit the same café 2 days in a row?
- (ii) What is the probability that you visit one of the cafés exactly once?

i. $\frac{1}{16}$

ii. $\frac{5}{16}$

EXERCISE 4. Consider a family that has two children. Our outcome space is

$$\{(G, G), (G, B), (B, G), (B, B)\}.$$

Assume that all four possible outcomes are equally likely.

- (i) What is the probability that both children are girls given that the first child is a girl?
- (ii) We ask the father: “Do you have at least one daughter?” He responds “Yes!” Given this extra information, what is the probability that both children are girls? In other words, what is the probability that both children are girls given that we know at least one of them is a girl?

i. $\frac{1}{2}$

ii. $\frac{1}{3}$

EXERCISE 5. Toss a fair coin 3 times.

- (i) What is the probability of 3 heads, HHH ?
- (ii) What is the probability that you see exactly one heads?
- (iii) Given that you have observed *at least* one heads, what is the probability that you observe at least 2 heads?

i. $\frac{1}{8}$

ii. $\frac{3}{8}$

iii. $\frac{4}{7}$

EXERCISE 6. A box contains three coins: two regular coins and one fake two-headed coin (which always shows heads).

- (i) You pick a coin at random and toss it. What is the probability that it lands heads up?

- (ii) You pick a coin at random and toss it, and get heads. What is the probability that it is the two-headed coin?

i. $\frac{2}{3}$

ii. $\frac{1}{2}$

EXERCISE 7. A king prepares 1000 packages, 10 of which contains precious rare diamonds, while the rest contains deadly snakes. Since you are a prominent and wise scientist, you receive one of these packages and develop a test to identify whether the package contains a diamond or a snake. Your test can have two potential results: Diamond or Snake.

If the package actually contains a snake, your test has a 99% probability of correctly identifying that the package contains a snake. If the package contains a diamond, your test has 70% probability of correctly identifying that the package contains a diamond.

- (i) Given that your test is positive for diamond (negative for snake), what is the probability that the package actually contains a diamond?
- (ii) Given that your test is negative for diamond (positive for snake), what is the probability that the package actually contains a diamond?

The rest of the question is hard, do only if you want to. Hereafter you can assume test and the "Tsss" sound are independent conditional on what is inside the package.

Now you learn that some packages make an ominous "Tsssss" sound. Another fellow scientist says if there is a snake in the package, there is a 90% chance that "Tsssss" sound will come. However, if there is a diamond in the package, there is also a 20% probability that "Tssss" sound will come because some diamonds also make an ominous sound.

- (iii) Given that your test is positive for diamonds, but there is a "Tssss" sound coming from the package, what is the probability that the package actually contains a diamond?

i. $\frac{70}{169}$

ii. $\frac{3}{9831}$

iii. $\frac{140}{1031}$